

Chapter 8

Baire Spaces and Dimension Theory

In this chapter, we introduce a class of topological spaces called the *Baire spaces*. The defining condition for a Baire space is a bit complicated to state, but it is often useful in the applications, in both analysis and topology. Most of the spaces we have been studying are Baire spaces. For instance, a Hausdorff space is a Baire space if it is compact, or even locally compact. And a metrizable space X is a Baire space if it is topologically complete, that is, if there is a metric for X relative to which X is complete.

It follows that, since the space $\mathcal{C}(X, \mathbb{R}^n)$ of all continuous functions from a space X to $\mathbb{R}^n$ is complete in the uniform metric, it is a Baire space in the uniform topology. This fact has a number of interesting applications.

One application is the proof we give in §49 of the existence of a continuous nowhere-differentiable real-valued function.

Another application arises in that branch of topology called *dimension theory*. In §50, we define a topological notion of dimension, due to Lebesgue. And we prove the classical theorem that every compact metrizable space of topological dimension m can be imbedded in euclidean space $\mathbb{R}^N$ of dimension $N = 2m + 1$. It follows that every compact m -manifold can be imbedded in $\mathbb{R}^{2m+1}$. This generalizes the imbedding theorem proved in §36.

Throughout the chapter, we assume familiarity with complete metric spaces (§43). When we study dimension theory, we shall make use of §36, Imbeddings of Manifolds, as well as a bit of linear algebra.

§48 Baire Spaces

The defining condition for a Baire space is probably as “unnatural looking” as any condition we have yet introduced in this book. But bear with us awhile.

In this section, we shall define Baire spaces and shall show that two important classes of spaces—the complete metric spaces and the compact Hausdorff spaces—are contained in the class of Baire spaces. Then we shall give some applications, which, even if they do not make the Baire condition seem any more natural, will at least show what a useful tool it can be. In fact, it turns out to be a very useful and fairly sophisticated tool in both analysis and topology.

Definition. Recall that if A is a subset of a space X , the *interior* of A is defined as the union of all open sets of X that are contained in A . To say that A has *empty interior* is to say then that A contains no open set of X other than the empty set. Equivalently, A has empty interior if every point of A is a limit point of the complement of A , that is, if the complement of A is dense in X .

EXAMPLE 1 The set $\mathbb{Q}$ of rationals has empty interior as a subset of $\mathbb{R}$, but the interval $[0, 1]$ has nonempty interior. The interval $[0, 1] \times 0$ has empty interior as a subset of the plane $\mathbb{R}^2$, and so does the subset $\mathbb{Q} \times \mathbb{R}$.

Definition. A space X is said to be a *Baire space* if the following condition holds: Given any countable collection $\{A_n\}$ of closed sets of X each of which has empty interior in X , their union $\bigcup A_n$ also has empty interior in X .

EXAMPLE 2 The space $\mathbb{Q}$ of rationals is not a Baire space. For each one-point set in $\mathbb{Q}$ is closed and has empty interior in $\mathbb{Q}$, and $\mathbb{Q}$ is the countable union of its one-point subsets.

The space $\mathbb{Z}_+$, on the other hand, does form a Baire space. Every subset of $\mathbb{Z}_+$ is open, so that there exist no subsets of $\mathbb{Z}_+$ having empty interior, except for the empty set. Therefore, $\mathbb{Z}_+$ satisfies the Baire condition vacuously.

More generally, every closed subspace of $\mathbb{R}$, being a complete metric space, is a Baire space. Somewhat surprising is the fact that the irrationals in $\mathbb{R}$ also form a Baire space; see Exercise 6.

The terminology originally used by R. Baire for this concept involved the word “category.” A subset A of a space X was said to be of the *first category* in X if it was contained in the union of a countable collection of closed sets of X having empty interiors in X ; otherwise, it was said to be of the *second category* in X . Using this terminology, we can say the following:

A space X is a Baire space if and only if every nonempty open set in X is of the second category.

We shall not use the terms “first category” and “second category” in this book.

The preceding definition is the “closed set definition” of a Baire space. There is also a formulation involving open sets that is frequently useful. It is given in the following lemma.

Lemma 48.1. *X is a Baire space if and only if given any countable collection $\{U_n\}$ of open sets in X , each of which is dense in X , their intersection $\bigcap U_n$ is also dense in X .*

Proof. Recall that a set C is dense in X if $\bar{C} = X$. The theorem now follows at once from the two remarks:

- (1) A is closed in X if and only if $X - A$ is open in X .
- (2) B has empty interior in X if and only if $X - B$ is dense in X . ■

There are a number of theorems giving conditions under which a space is a Baire space. The most important is the following:

Theorem 48.2 (Baire category theorem). *If X is a compact Hausdorff space or a complete metric space, then X is a Baire space.*

Proof. Given a countable collection $\{A_n\}$ of closed set of X having empty interiors, we want to show that their union $\bigcup A_n$ also has empty interior in X . So, given the nonempty open set U_0 of X , we must find a point x of U_0 that does not lie in any of the sets A_n .

Consider the first set A_1 . By hypothesis, A_1 does not contain U_0 . Therefore, we may choose a point y of U_0 that is not in A_1 . Regularity of X , along with the fact that A_1 is closed, enables us to choose a neighborhood U_1 of y such that

$$\begin{aligned}\bar{U}_1 \cap A_1 &= \emptyset, \\ \bar{U}_1 &\subset U_0.\end{aligned}$$

If X is metric, we also choose U_1 small enough that its diameter is less than 1.

In general, given the nonempty open set U_{n-1} , we choose a point of U_{n-1} that is not in the closed set A_n , and then we choose U_n to be a neighborhood of this point such that

$$\begin{aligned}\bar{U}_n \cap A_n &= \emptyset, \\ \bar{U}_n &\subset U_{n-1}, \\ \text{diam } U_n &< 1/n \quad \text{in the metric case.}\end{aligned}$$

We assert that the intersection $\bigcap \bar{U}_n$ is nonempty. From this fact, our theorem will follow. For if x is a point of $\bigcap \bar{U}_n$, then x is in U_0 because $\bar{U}_1 \subset U_0$. And for each n , the point x is not in A_n because $\bar{U}_n$ is disjoint from A_n .

The proof that $\bigcap \bar{U}_n$ is nonempty splits into two parts, depending on whether X is compact Hausdorff or complete metric. If X is compact Hausdorff, we consider the nested sequence $\bar{U}_1 \supset \bar{U}_2 \supset \cdots$ of nonempty subsets of X . The collection $\{\bar{U}_n\}$ has the finite intersection property; since X is compact, the intersection $\bigcap \bar{U}_n$ must be nonempty.

If X is complete metric, we apply the following lemma. ■

Lemma 48.3. Let $C_1 \supset C_2 \supset \cdots$ be a nested sequence of nonempty closed sets in the complete metric space X . If $\text{diam } C_n \rightarrow 0$, then $\bigcap C_n \neq \emptyset$.

Proof. We gave this as an exercise in §43. Here is a proof: Choose $x_n \in C_n$ for each n . Because $x_n, x_m \in C_N$ for $n, m \geq N$, and because $\text{diam } C_N$ can be made less than any given ϵ by choosing N large enough, the sequence (x_n) is a Cauchy sequence. Suppose that it converges to x . Then for given k , the subsequence $x_k, x_{k+1}, \dots$ also converges to x . Thus x necessarily belongs to $\bar{C}_k = C_k$. Then $x \in \bigcap C_k$, as desired. ■

Here is one application of the theory of Baire spaces; we shall give further applications in the sections that follow. This application is perhaps more amusing than profound. It concerns a question that a student might ask concerning convergent sequences of continuous functions.

Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of continuous functions such that $f_n(x) \rightarrow f(x)$ for each $x \in [0, 1]$. There are examples that show the limit function f need not be continuous. But one might wonder just how discontinuous f can be. Could it be discontinuous everywhere, for instance? The answer is "no." We shall show that f must be continuous at infinitely many points of $[0, 1]$. In fact, the set of points at which f is continuous is dense in $[0, 1]$!

To prove this result, we need the following lemma:

***Lemma 48.4.** Any open subspace Y of a Baire space X is itself a Baire space.

Proof. Let A_n be a countable collection of closed sets of Y that have empty interiors in Y . We show that $\bigcup A_n$ has empty interior in Y .

Let $\bar{A}_n$ be the closure of A_n in X ; then $\bar{A}_n \cap Y = A_n$. The set $\bar{A}_n$ has empty interior in X . For if U is a nonempty open set of X contained in $\bar{A}_n$, then U must intersect A_n . Then $U \cap Y$ is a nonempty open set of Y contained in A_n , contrary to hypothesis.

If the union of the sets A_n contains the nonempty open set W of Y , then the union of the sets $\bar{A}_n$ also contains the set W , which is open in X because Y is open in X . But each set $\bar{A}_n$ has empty interior in X , contradicting the fact that X is a Baire space. ■

***Theorem 48.5.** Let X be a space; let (Y, d) be a metric space. Let $f_n : X \rightarrow Y$ be a sequence of continuous functions such that $f_n(x) \rightarrow f(x)$ for all $x \in X$, where $f : X \rightarrow Y$. If X is a Baire space, the set of points at which f is continuous is dense in X .

Proof. Given a positive integer N and given $\epsilon > 0$, define

$$A_N(\epsilon) = \{x \mid d(f_n(x), f_m(x)) \leq \epsilon \text{ for all } n, m \geq N\}.$$

Note that $A_N(\epsilon)$ is closed in X . For the set of those x for which $d(f_n(x), f_m(x)) \leq \epsilon$ is closed in X , by continuity of f_n and f_m , and $A_N(\epsilon)$ is the intersection of these sets for all $n, m \geq N$.

For fixed ϵ , consider the sets $A_1(\epsilon) \subset A_2(\epsilon) \subset \cdots$. The union of these sets is all of X . For, given $x_0 \in X$, the fact that $f_n(x_0) \rightarrow f(x_0)$ implies that the sequence $f_n(x_0)$ is a Cauchy sequence; hence $x_0 \in A_N(\epsilon)$ for some N .

Now let

$$U(\epsilon) = \bigcup_{N \in \mathbb{Z}_+} \text{Int } A_N(\epsilon).$$

We shall prove two things:

- (1) $U(\epsilon)$ is open and dense in X .
- (2) The function f is continuous at each point of the set

$$C = U(1) \cap U(1/2) \cap U(1/3) \cap \cdots.$$

Our theorem then follows from the fact that X is a Baire space.

To show that $U(\epsilon)$ is dense in X , it suffices to show that for any nonempty open set V of X , there is an N such that the set $V \cap \text{Int } A_N(\epsilon)$ is nonempty. For this purpose, we note first that for each N , the set $V \cap A_N(\epsilon)$ is closed in V . Because V is a Baire space by the preceding lemma, at least one of these sets, say $V \cap A_M(\epsilon)$, must contain a nonempty open set W of V . Because V is open in X , the set W is open in X ; therefore, it is contained in $\text{Int } A_M(\epsilon)$.

Now we show that if $x_0 \in C$, then f is continuous at x_0 . Given $\epsilon > 0$, we shall find a neighborhood W of x_0 such that $d(f(x), f(x_0)) < \epsilon$ for $x \in W$.

First, choose k so that $1/k < \epsilon/3$. Since $x_0 \in C$, we have $x_0 \in U(1/k)$; therefore, there is an N such that $x_0 \in \text{Int } A_N(1/k)$. Finally, continuity of the function f_N enables us to choose a neighborhood W of x_0 , contained in $A_N(1/k)$, such that

$$(*) \quad d(f_N(x), f_N(x_0)) < \epsilon/3 \quad \text{for } x \in W.$$

The fact that $W \subset A_N(1/k)$ implies that

$$d(f_n(x), f_n(x)) \leq 1/k \quad \text{for } n \geq N \text{ and } x \in W.$$

Letting $n \rightarrow \infty$, we obtain the inequality

$$(**) \quad d(f(x), f_N(x)) \leq 1/k < \epsilon/3 \quad \text{for } x \in W.$$

In particular, since $x_0 \in W$, we have

$$(***) \quad d(f(x_0), f_N(x_0)) < \epsilon/3.$$

Applying the triangle inequality to (*), (**), and (***) gives us our desired result. ■

Exercises

1. Let X equal the countable union $\bigcup B_n$. Show that if X is a nonempty Baire space, at least one of the sets $\bar{B}_n$ has a nonempty interior.

2. The Baire category theorem implies that $\mathbb{R}$ cannot be written as a countable union of closed subsets having empty interiors. Show this fails if the sets are not required to be closed.
3. Show that every locally compact Hausdorff space is a Baire space.
4. Show that if every point x of X has a neighborhood that is a Baire space, then X is a Baire space. [Hint: Use the open set formulation of the Baire condition.]
5. Show that if Y is a G_δ set in X , and if X is compact Hausdorff or complete metric, then Y is a Baire space in the subspace topology. [Hint: Suppose that $Y = \bigcap W_n$, where W_n is open in X , and that B_n is closed in Y and has empty interior in Y . Given U_0 open in X with $U_0 \cap Y \neq \emptyset$, find a sequence of open sets U_n of X with $U_n \cap Y$ nonempty, such that

$$\begin{aligned}\bar{U}_n &\subset U_{n-1}, \\ \bar{U}_n \cap \bar{B}_n &= \emptyset, \\ \text{diam } U_n &< 1/n \quad \text{in the metric case,} \\ \bar{U}_n &\subset W_n.\end{aligned}$$

6. Show that the irrationals are a Baire space.
7. Prove the following:

Theorem. If D is a countable dense subset of $\mathbb{R}$, there is no function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous precisely at the points of D .

Proof.

- (a) Show that if $f : \mathbb{R} \rightarrow \mathbb{R}$, then the set C of points at which f is continuous is a G_δ set in $\mathbb{R}$. [Hint: Let U_n be the union of all open sets U of $\mathbb{R}$ such that $\text{diam } f(U) < 1/n$. Show that $C = \bigcap U_n$.]
- (b) Show that D is not a G_δ set in $\mathbb{R}$. [Hint: Suppose $D = \bigcap W_n$, where W_n is open in $\mathbb{R}$. For $d \in D$, set $V_d = \mathbb{R} - \{d\}$. Show W_n and V_d are dense in $\mathbb{R}$.]
8. If f_n is a sequence of continuous functions $f_n : \mathbb{R} \rightarrow \mathbb{R}$ such that $f_n(x) \rightarrow f(x)$ for each $x \in \mathbb{R}$, show that f is continuous at uncountably many points of $\mathbb{R}$.
9. Let $g : \mathbb{Z}_+ \rightarrow \mathbb{Q}$ be a bijective function; let $x_n = g(n)$. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ as follows:

$$\begin{aligned}f(x_n) &= 1/n & \text{for } x_n \in \mathbb{Q}, \\ f(x) &= 0 & \text{for } x \notin \mathbb{Q}\end{aligned}$$

Show that f is continuous at each irrational and discontinuous at each rational. Can you find a sequence of continuous functions f_n converging to f ?

10. Prove the following:

Theorem (Uniform boundedness principle). Let X be a complete metric space, and let $\mathcal{F}$ be a subset of $\mathcal{C}(X, \mathbb{R})$ such that for each $a \in X$, the set

$$\mathcal{F}_a = \{f(a) \mid f \in \mathcal{F}\}$$

is bounded. Then there is a nonempty open set U of X on which the functions in $\mathcal{F}$ are uniformly bounded, that is, there is a number M such that $|f(x)| \leq M$ for all $x \in U$ and all $f \in \mathcal{F}$. [Hint: Let $A_N = \{x; |f(x)| \leq N \text{ for all } f \in \mathcal{F}\}$.]

11. Determine whether or not $\mathbb{R}_\ell$ is a Baire space
12. Show that $\mathbb{R}^J$ is a Baire space in the box, product, and uniform topologies.
- *13. Let X be a topological space; let Y be a complete metric space. Show that $\mathcal{C}(X, Y)$ is a Baire space in the fine topology (see Exercise 11 of §46). [Hint: Given basis elements $B(f_i, \delta_i)$ such that $\delta_i \leq 1$ and $\delta_{i+1} \leq \delta_i/3$ and $f_{i+1} \in B(f_i, \delta_i/3)$, show that

$$\bigcap B(f_i, \delta_i) \neq \emptyset.]$$

*§49 A Nowhere-Differentiable Function

We prove the following result from analysis:

Theorem 49.1. Let $h : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Given $\epsilon > 0$, there is a function $g : [0, 1] \rightarrow \mathbb{R}$ with $|h(x) - g(x)| < \epsilon$ for all x , such that g is continuous and nowhere differentiable.

Proof. Let $I = [0, 1]$. Consider the space $\mathcal{C} = \mathcal{C}(I, \mathbb{R})$ of continuous maps from I to $\mathbb{R}$, in the metric

$$\rho(f, g) = \max\{|f(x) - g(x)|\}.$$

This space is a complete metric space and, therefore, is a Baire space. We shall define, for each n , a certain subset U_n of $\mathcal{C}$ that is open in $\mathcal{C}$ and dense in $\mathcal{C}$, and has the property that the functions belonging to the intersection

$$\bigcap_{n \in \mathbb{Z}_+} U_n$$

are nowhere differentiable. Because $\mathcal{C}$ is a Baire space, this intersection is dense in $\mathcal{C}$, by Lemma 48.1. Therefore, given h and ϵ , this intersection must contain a function g such that $\rho(h, g) < \epsilon$. The theorem follows.

The tricky part is to define the set U_n properly. We first take a function f and consider its difference quotients. Given $x \in I$ and given $0 < h \leq \frac{1}{2}$, consider the expressions

$$\left| \frac{f(x+h) - f(x)}{h} \right| \quad \text{and} \quad \left| \frac{f(x-h) - f(x)}{-h} \right|.$$

Since $h \leq \frac{1}{2}$, at least one of the numbers $x+h$ and $x-h$ belongs to I , so that at least one of these expressions is defined. Let $\Delta f(x, h)$ denote the larger of the two if both

are defined; otherwise, let it denote the one that is defined. If the derivative $f'(x)$ of f at x exists, it equals the limit of these difference quotients, so that

$$|f'(x)| = \lim_{h \rightarrow 0} \Delta f(x, h).$$

We seek to find a continuous function for which this limit does not exist. To be specific, we shall construct f so that given x , there is a sequence of numbers h_n converging to 0 for which the numbers $\Delta f(x, h_n)$ become arbitrarily large.

This gives us the idea for defining the set U_n . Given any positive number $h \leq 1/2$, let

$$\Delta_h f = \inf\{\Delta f(x, h) \mid x \in I\}$$

Then for $n \geq 2$, we define U_n by declaring that a function f belongs to U_n if and only if for some positive number $h \leq 1/n$, we have $\Delta_h f > n$.

EXAMPLE 1 Let $\alpha > 0$ be given. The function $f: [0, 1] \rightarrow \mathbb{R}$ given by the equation $f(x) = 4\alpha x(1-x)$, whose graph is a parabola, satisfies the condition $\Delta f(x, h) \geq \alpha$ for $h = 1/4$ and all x , as you can check. Geometrically speaking, what this says is that for each x , at least one of the indicated secant lines of the parabola in Figure 49.1 has slope of absolute value at least α . Hence if $\alpha > 4$, the function f belongs to U_4 . The function g pictured in Figure 49.1 satisfies the condition $\Delta g(x, h) \geq \alpha$ for any $h \leq 1/4$; hence g belongs to U_n provided $\alpha > n$. The function k satisfies the condition $k(x, h) \geq \alpha$ for any $h \leq 1/8$; hence k belongs to U_n if $\alpha > n$.

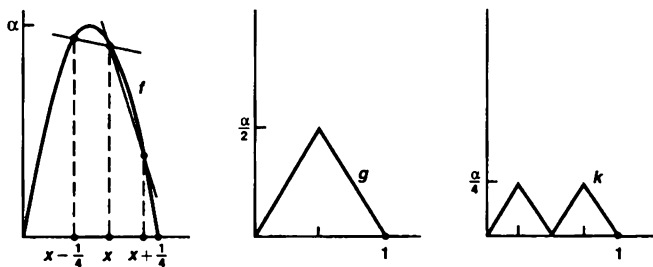


Figure 49.1

Now we prove the following facts about the set U_n :

(1) $\bigcap U_n$ consists of nowhere-differentiable functions. Let $f \in \bigcap U_n$. We shall prove that given x in $[0, 1]$, the limit

$$\lim \Delta f(x, h)$$

does not exist: Given n , the fact that f belongs to U_n means that we can find a number h_n with $0 < h_n \leq 1/n$ such that

$$\Delta f(x, h_n) > n.$$

Then the sequence (h_n) converges to zero, but the sequence $(\Delta f(x, h_n))$ does not converge. As a result, f is not differentiable at x .

(2) U_n is open in $\mathcal{C}$. Suppose that $f \in U_n$; we find a δ -neighborhood of f that is contained in U_n . Because $f \in U_n$, there is a number h with $0 < h \leq 1/n$ such that $\Delta_h f > n$. Set $M = \Delta_h f$, and let

$$\delta = h(M - n)/4.$$

We assert that if g is a function with $\rho(f, g) < \delta$, then

$$\Delta g(x, h) \geq \frac{1}{2}(M + n) > n$$

for all $x \in I$, so that $g \in U_n$.

To prove the assertion, let us first assume that $\Delta f(x, h)$ is equal to the quotient $|f(x+h) - f(x)|/h$. We compute

$$\left| \frac{f(x+h) - f(x)}{h} - \frac{g(x+h) - g(x)}{h} \right| = \\ (1/h)|[f(x+h) - g(x+h)] - [f(x) - g(x)]| \leq 2\delta/h = (M - n)/2.$$

If the first difference quotient is at least M in absolute value, then the second is in absolute value at least

$$M - \frac{1}{2}(M - n) = \frac{1}{2}(M + n).$$

A similar remark applies if $\Delta f(x, h)$ equals the other difference quotient.

(3) U_n is dense in $\mathcal{C}$. We must show that given f in $\mathcal{C}$, given $\epsilon > 0$, and given n , we can find an element g of U_n within ϵ of f .

Choose $\alpha > n$. We shall construct g as a "piecewise-linear" function, that is, a function whose graph is a broken line segment; each line segment in the graph of g will have slope at least α in absolute value. It follows at once that such a function g belongs to U_n . For let

$$0 = x_0 < x_1 < x_2 < \cdots < x_k = 1$$

be a partition of the interval $[0, 1]$ such that the restriction of g to each subinterval $I_i = [x_{i-1}, x_i]$ is a linear function. Then choose h so that $h \leq 1/n$ and

$$h \leq \frac{1}{2} \min\{|x_i - x_{i-1}|; i = 1, \dots, k\}.$$

If x is in $[0, 1]$, then x belongs to some subinterval I_i . If x belongs to the first half of the subinterval I_i , then $x+h$ belongs to I_i and $(g(x+h) - g(x))/h$ equals the slope of the linear function $g|_{I_i}$. Similarly, if x belongs to the second half of I_i , then $x-h$ belongs to I_i and $(g(x-h) - g(x))/(-h)$ equals the slope of $g|_{I_i}$. In either case, $\Delta g(x, h) \geq \alpha$, so $g \in U_n$, as desired.

Now given f , ϵ , and α , we must show how to construct the desired piecewise-linear function g . First, we use uniform continuity of f to choose a partition of the interval

$$0 = t_0 < t_1 < \cdots < t_m = 1$$

having the property that f varies by at most $\epsilon/4$ on each subinterval $[t_{i-1}, t_i]$ of this partition. For each $i = 1, \dots, m$, choose a point $a_i \in (t_{i-1}, t_i)$. We then define a piecewise-linear function g_1 by the equations

$$g_1(x) = \begin{cases} f(t_{i-1}) & \text{for } x \in [t_{i-1}, a_i], \\ f(t_{i-1}) + m_i(x - a_i) & \text{for } x \in [a_i, t_i], \end{cases}$$

where $m_i = (f(t_i) - f(t_{i-1})) / (t_i - a_i)$. The graphs of f and g_1 are pictured in Figure 49.2.

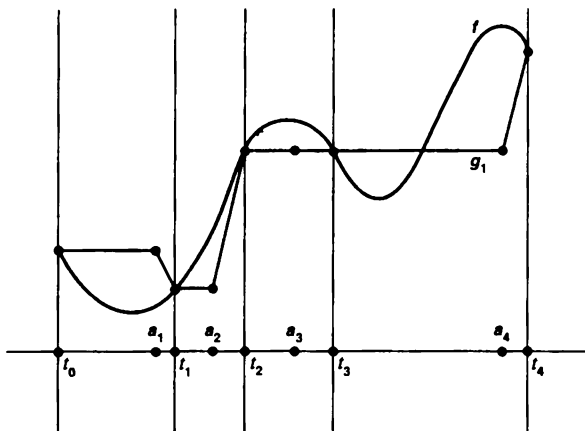


Figure 49.2

We have some freedom of choice in choosing the point a_i . If $f(t_i) \neq f(t_{i-1})$, we require a_i to be close enough to t_i that

$$t_i - a_i < \frac{|f(t_i) - f(t_{i-1})|}{\alpha}$$

Then the graph of g_1 will consist entirely of line segments of slope zero and line segments of slope at least α in absolute value.

Furthermore, we assert that $\rho(g_1, f) \leq \epsilon/2$. On the interval I_i , both $g_1(x)$ and $f(x)$ vary by at most $\epsilon/4$ from $f(t_{i-1})$; therefore, they are within $\epsilon/2$ of each other. Then $\rho(g_1, f) = \max\{|g_1(x) - f(x)|\} \leq \epsilon/2$.

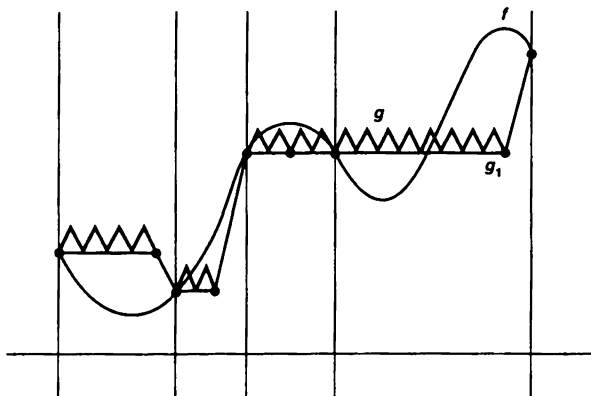


Figure 49.3

The function g_1 is not yet the function we want. We now define a function g by replacing each horizontal line segment in the graph of g_1 by a “sawtooth” graph that lies within $\epsilon/2$ of the graph of g_1 and has the property that each edge of the sawtooth has slope at least α in absolute value. We leave this part of the construction to you. The result is the desired piecewise-linear function g . See Figure 49.3. ■

You may find this proof frustrating, in that it seems so abstract and nonconstructive. Implicit in the proof, however, is a procedure for constructing a specific sequence f_n of piecewise-linear functions that converges uniformly to the nowhere-differentiable function f . And defining the function f in this way is just as constructive as the usual definition of the sine function, for instance, as the limit of an infinite series.

Exercises

1. Check the stated properties of the functions f , g , and k of Example 1.
2. Given n and ϵ , define a continuous function $f : I \rightarrow \mathbb{R}$ such that $f \in U_n$ and $|f(x)| \leq \epsilon$ for all x .

§50 Introduction to Dimension Theory

We showed in §36 that if X is a compact manifold, then X can be imbedded in $\mathbb{R}^N$ for some positive integer N . In this section, we generalize this theorem to arbitrary compact metrizable spaces

We shall define, for an arbitrary topological space X , a notion of topological dimension. It is the "covering dimension" originally defined by Lebesgue. We shall prove that each compact subset of $\mathbb{R}^m$ has topological dimension at most m . We shall also prove that the topological dimension of any compact m -manifold is at most m . (It is, in fact, precisely m , but this we shall not prove.)

The major theorem of this section is the theorem, due to K. Menger and G. Nöbeling, that any compact metrizable space of topological dimension m can be imbedded in $\mathbb{R}^N$ for $N = 2m + 1$. The proof is an application of the Baire theorem. It follows that every compact m -manifold can be imbedded in $\mathbb{R}^{2m+1}$. It follows also that a compact metrizable space can be imbedded in $\mathbb{R}^N$ for some N if and only if it has finite topological dimension.

Much of what we shall do holds without requiring the space in question to be compact. But we shall restrict ourselves to that case whenever it is convenient to do so. Generalizations to the noncompact case are given in the exercises.

Definition. A collection $\mathcal{A}$ of subsets of the space X is said to have *order* $m + 1$ if some point of X lies in $m + 1$ elements of $\mathcal{A}$, and no point of X lies in more than $m + 1$ elements of $\mathcal{A}$.

Now we define what we mean by the *topological dimension* of a space X . Recall that given a collection $\mathcal{A}$ of subsets of X , a collection $\mathcal{B}$ is said to *refine* $\mathcal{A}$, or to be a *refinement* of $\mathcal{A}$, if for each element B of $\mathcal{B}$ there is an element A of $\mathcal{A}$ such that $B \subset A$.

Definition. A space X is said to be *finite dimensional* if there is some integer m such that for every open covering $\mathcal{A}$ of X , there is an open covering $\mathcal{B}$ of X that refines $\mathcal{A}$ and has order at most $m + 1$. The *topological dimension* of X is defined to be the smallest value of m for which this statement holds; we denote it by $\dim X$.

EXAMPLE 1. Any compact subspace X of $\mathbb{R}$ has topological dimension at most 1. We begin by defining an open covering of $\mathbb{R}$ of order 2. Let $\mathcal{A}_1$ denote the collection of all open intervals of the form $(n, n + 1)$ in $\mathbb{R}$, where n is an integer. Let $\mathcal{A}_0$ denote the collection of all open intervals of the form $(n - 1/2, n + 1/2)$, for n an integer. Then $\mathcal{A} = \mathcal{A}_0 \cup \mathcal{A}_1$ is an open covering of $\mathbb{R}$ by sets of diameter one. Because no two elements of $\mathcal{A}_0$ intersect, and no two elements of $\mathcal{A}_1$ intersect, $\mathcal{A}$ has order 2.

Now let X be a compact subspace of $\mathbb{R}$. Given a covering $\mathcal{C}$ of X by sets open in X , this covering has a positive Lebesgue number δ . This means that any collection of subsets of X that have diameter less than δ is automatically a refinement of $\mathcal{C}$. Consider the homeomorphism $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = (\frac{1}{2}\delta)x$. The images under f of the elements of the collection $\mathcal{A}$ form an open covering of $\mathbb{R}$ of order 2 whose elements have diameter $\frac{1}{2}\delta$; their intersections with X form the required open covering of X .

EXAMPLE 2. The interval $X = [0, 1]$ has topological dimension 1. We know that $\dim X \leq 1$. To show equality holds, let $\mathcal{A}$ be the covering of X by the sets $[0, 1)$ and $(0, 1]$. We show that if $\mathcal{B}$ is any open covering of X that refines $\mathcal{A}$, then $\mathcal{B}$ has order at least 2. Since $\mathcal{B}$ refines $\mathcal{A}$, it must contain more than one element. Let U be one of the elements

of $\mathcal{B}$ and let V be the union of the others. If $\mathcal{B}$ had order 1, then the sets U and V would be disjoint and would thus form a separation of X . We conclude that $\mathcal{B}$ has order at least 2.

EXAMPLE 3. Any compact subspace X of $\mathbb{R}^2$ has topological dimension at most 2. To prove this fact, we construct a certain open covering $\mathcal{A}$ of $\mathbb{R}^2$ that has order 3. We begin by defining $\mathcal{A}_2$ to be the collection of all open unit squares in $\mathbb{R}^2$ of the following form:

$$\mathcal{A}_2 = \{(n, n+1) \times (m, m+1) \mid n, m \text{ integers}\}$$

Note that the elements of $\mathcal{A}_2$ are disjoint. Then, we define a collection $\mathcal{A}_1$ by taking each (open) edge e of one of these squares,

$$e = (n) \times (m, m+1) \quad \text{or} \quad e = (n, n+1) \times (m),$$

and expanding it slightly to an open set U_e of $\mathbb{R}^2$, being careful to ensure that if $e \neq e'$, the sets U_e and $U_{e'}$ are disjoint. We also choose each U_e so that its diameter is at most 2. Finally, we define $\mathcal{A}_0$ to be the collection consisting of all open balls of radius $\frac{1}{2}$ about the points $n \times m$. See Figure 50.1

The collection of open sets $\mathcal{A} = \mathcal{A}_2 \cup \mathcal{A}_1 \cup \mathcal{A}_0$ covers $\mathbb{R}^2$. Each of its elements has diameter at most 2. And it has order 3, since no point of $\mathbb{R}^2$ can lie in more than one set from each $\mathcal{A}_i$.

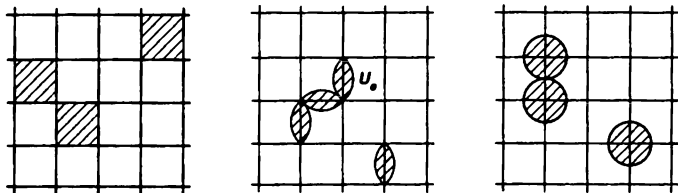


Figure 50.1

Now let X be a compact subspace of $\mathbb{R}^2$. Given an open covering of X , it has a positive Lebesgue number δ . Consider the homeomorphism $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by the equation $f(x) = (\delta/3)x$. The images under f of the open sets of the collection $\mathcal{A}$ form an open covering of $\mathbb{R}^2$ by sets of diameter less than δ , their intersections with X form the required open covering of X .

We shall generalize this result to compact subsets of $\mathbb{R}^n$ shortly.

Some basic facts about topological dimension are given in the following theorems:

Theorem 50.1. Let X be a space having finite dimension. If Y is a closed subspace of X , then Y has finite dimension and $\dim Y \leq \dim X$.

Proof. Let $\dim X = m$. Let $\mathcal{A}$ be a covering of Y by sets open in Y . For each $A \in \mathcal{A}$, choose an open set A' of X such that $A' \cap Y = A$. Cover X by the open sets A' , along with the open set $X - Y$. Let $\mathcal{B}$ be a refinement of this covering that is an open covering of X and has order at most $m + 1$. Then the collection

$$\{B \cap Y \mid B \in \mathcal{B}\}$$

is a covering of Y by sets open in Y , it has order at most $m + 1$, and it refines $\mathcal{A}$. ■

Theorem 50.2. Let $X = Y \cup Z$, where Y and Z are closed subspaces of X having finite topological dimension. Then

$$\dim X = \max\{\dim Y, \dim Z\}.$$

Proof. Let $m = \max\{\dim Y, \dim Z\}$. We shall show that X is finite dimensional and has topological dimension at most m . It then follows from the preceding theorem that X has topological dimension precisely m .

Step 1. If $\mathcal{A}$ is an open covering of X , we say that $\mathcal{A}$ has order at most $m + 1$ at points of Y provided no point of Y lies in more than $m + 1$ elements of $\mathcal{A}$.

We show that if $\mathcal{A}$ is an open covering of X , then there is an open covering of X that refines $\mathcal{A}$ and has order at most $m + 1$ at points of Y .

To prove this fact, consider the collection

$$\{A \cap Y \mid A \in \mathcal{A}\}.$$

It is an open covering of Y , so it has a refinement $\mathcal{B}$ that is an open covering of Y and has order at most $m + 1$. Given $B \in \mathcal{B}$, choose an open set U_B of X such that $U_B \cap Y = B$. Choose also an element A_B of $\mathcal{A}$ such that $B \subset A_B$. Let $\mathcal{C}$ be the collection consisting of all the sets $U_B \cap A_B$, for $B \in \mathcal{B}$, along with all the sets $A - Y$, for $A \in \mathcal{A}$. Then $\mathcal{C}$ is the desired open covering of X .

Step 2. Now let $\mathcal{A}$ be an open covering of X . We construct an open covering $\mathcal{D}$ of X that refines $\mathcal{A}$ and has order at most $m + 1$. Let $\mathcal{B}$ be an open covering of X refining $\mathcal{A}$ that has order at most $m + 1$ at points of Y . Then let $\mathcal{C}$ be an open covering of X refining $\mathcal{B}$ that has order at most $m + 1$ at points of Z . •

We form a new covering $\mathcal{D}$ of X as follows: Define $f: \mathcal{C} \rightarrow \mathcal{B}$ by choosing for each $C \in \mathcal{C}$ an element $f(C)$ of $\mathcal{B}$ such that $C \subset f(C)$. Given $B \in \mathcal{B}$, define $D(B)$ to be the union of all those elements C of $\mathcal{C}$ for which $f(C) = B$. (Of course, $D(B)$ is empty if B is not in the image of f .) Let $\mathcal{D}$ be the collection of all the sets $D(B)$, for $B \in \mathcal{B}$.

Now $\mathcal{D}$ refines $\mathcal{B}$, because $D(B) \subset B$ for each B ; therefore, $\mathcal{D}$ refines $\mathcal{A}$. Also, $\mathcal{D}$ covers X because $\mathcal{C}$ covers X and $C \subset D(f(C))$ for each $C \in \mathcal{C}$. We show that $\mathcal{D}$ has order at most $m + 1$. Suppose $x \in D(B_1) \cap \cdots \cap D(B_k)$, where the sets $D(B_i)$ are distinct. We wish to prove that $k \leq m + 1$. Note that the sets $B_1, \dots, B_k$ must be distinct because the sets $D(B_i)$ are. Because $x \in D(B_i)$, we can choose for each i , a set $C_i \in \mathcal{C}$ such that $x \in C_i$ and $f(C_i) = B_i$. The sets C_i are distinct because the sets B_i are. Furthermore,

$$x \in [C_1 \cap \cdots \cap C_k] \subset [D(B_1) \cap \cdots \cap D(B_k)] \subset [B_1 \cap \cdots \cap B_k].$$

If x happens to lie in Y , then $k \leq m + 1$ because $\mathcal{B}$ has order at most $m + 1$ at points of Y ; and if x is in Z , then $k \leq m + 1$ because $\mathcal{C}$ has order at most $m + 1$ at points of Z . ■

Corollary 50.3. Let $X = Y_1 \cup \cdots \cup Y_k$, where each Y_i is a closed subspace of X and is finite dimensional. Then

$$\dim X = \max\{\dim Y_1, \dots, \dim Y_k\}.$$

EXAMPLE 4. Every compact 1-manifold X has topological dimension 1. The space X can be written as a finite union of spaces that are homeomorphic to the unit interval $[0, 1]$; then the preceding corollary applies

EXAMPLE 5. Every compact 2-manifold X has topological dimension at most 2. The space X can be written as a finite union of spaces that are homeomorphic to the closed unit ball in $\mathbb{R}^2$, then the preceding corollary applies.

An obvious question occurs at this point: Does a compact 2-manifold have topological dimension precisely 2? The answer is "yes," but the proof is not easy; it requires the tools of algebraic topology. We will prove in Part II of this book that every closed triangular region in $\mathbb{R}^2$ has topological dimension at least 2. (See §55.) It then follows that any compact subspace of $\mathbb{R}^2$ that contains a closed triangular region has topological dimension 2, from which it follows that every compact 2-manifold has topological dimension 2

EXAMPLE 6. An *arc* A is a space homeomorphic to the closed unit interval; the *end points* of A are the points p and q such that $A - \{p\}$ and $A - \{q\}$ are connected. A (finite) *linear graph* G is a Hausdorff space that is written as the union of finitely many arcs, each pair of which intersect in at most a common end point. The arcs in the collection are called the *edges* of G , and the end points of the arcs are called the *vertices* of G . Each edge of G , being compact, is closed in G ; the preceding corollary tells us that G has topological dimension 1

Two particular linear graphs are sketched in Figure 50.2. The first is a diagram of the familiar "gas-water-electricity problem"; the second is called the "complete graph on five vertices." Neither of them can be imbedded in $\mathbb{R}^2$. Although this fact is "intuitively obvious," it is highly nontrivial to prove. We shall give a proof in §64.

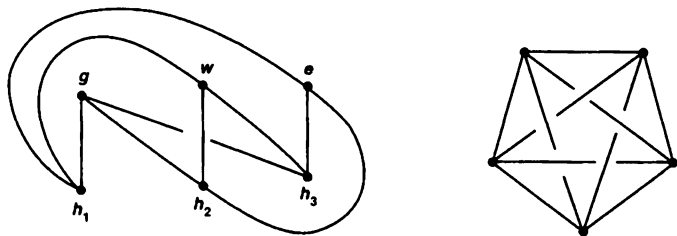


Figure 50.2

EXAMPLE 7. Every finite linear graph can be imbedded in $\mathbb{R}^3$. The proof involves the notion of "general position." A set S of points of $\mathbb{R}^3$ is said to be in *general position* if no three of the points of S are collinear and no four of them are coplanar. It is easy to find such a set of points. For example, the points of the curve

$$S = \{(t, t^2, t^3) \mid t \in \mathbb{R}\}$$

are in general position. For if four of these points belonged to a single plane $Ax + By + Cz = D$, then the polynomial equation

$$At + Bt^2 + Ct^3 = D$$

would have four distinct real roots! And if three of these points belonged to a single line, we could take an additional point of S and obtain four points that lie on a plane.

Now, given a finite linear graph G , with vertices $v_1, \dots, v_n$, let us choose a set $\{z_1, \dots, z_n\}$ of points of $\mathbb{R}^3$ that is in general position. Define a map $f: G \rightarrow \mathbb{R}^3$ by letting f map the vertex v_i to the point z_i , and map the edge joining v_i and v_j homeomorphically onto the line segment joining z_i and z_j . Now each edge of G is closed in G . It follows that f is continuous, by the pasting lemma. We show that f is injective, from which it follows that f is an imbedding. Let $e = v_i v_j$ and $e' = v_k v_m$ be two edges of G . If they have no vertex in common, then the line segments $f(e)$ and $f(e')$ are disjoint, for otherwise the points z_i, z_j, z_k, z_m would be coplanar. And if e and e' have a vertex in common, so that $i = k$, say, then the line segments $f(e)$ and $f(e')$ intersect only in the point $z_i = z_k$, for otherwise z_i, z_j , and z_m would be collinear.

Now we prove our general imbedding theorem, to the effect that every compact metrizable space of topological dimension m can be imbedded in $\mathbb{R}^{2m+1}$. This theorem is another "deep" theorem; it is not at all obvious, for instance, why $2m+1$ should be the crucial dimension. That will come out in the course of the proof.

To prove the imbedding theorem, we shall need to generalize the notion of general position to $\mathbb{R}^N$. This involves a bit of the analytic geometry of $\mathbb{R}^N$, which is nothing more than the usual linear algebra of $\mathbb{R}^N$ translated into somewhat different language.

Definition. A set $\{x_0, \dots, x_k\}$ of points of $\mathbb{R}^N$ is said to be *geometrically independent*, or *affinely independent*, if the equations

$$\sum_{i=0}^k a_i x_i = 0 \quad \text{and} \quad \sum_{i=0}^k a_i = 0$$

hold only if each $a_i = 0$.

Obviously, a set consisting of only one point is geometrically independent. But what does geometric independence mean in general? If we solve the second equation for a_0 and plug the answer into the first equation, we see that this definition is equivalent to the statement that the equation

$$\sum_{i=1}^k a_i (x_i - x_0) = 0$$

holds only if each $a_i = 0$. This is just the definition of *linear independence* for the set of vectors $x_1 - x_0, \dots, x_k - x_0$ of the vector space $\mathbb{R}^N$. This gives us something to visualize: Any two distinct points form a geometrically independent set. Three points form a geometrically independent set if they are not collinear. Four points in $\mathbb{R}^3$ form a geometrically independent set if they are not coplanar. And so on.

It follows from these remarks that the points

$$\begin{aligned} \mathbf{0} &= (0, 0, \dots, 0), \\ \epsilon_1 &= (1, 0, \dots, 0), \\ &\vdots \\ \epsilon_N &= (0, 0, \dots, 1) \end{aligned}$$

are geometrically independent in $\mathbb{R}^N$. It also follows that any geometrically independent set of points in $\mathbb{R}^N$ contains no more than $N + 1$ points

Definition. Let $\{\mathbf{x}_0, \dots, \mathbf{x}_k\}$ be a set of points of $\mathbb{R}^N$ that is geometrically independent. The *plane P determined by these points* is defined to be the set of all points $\mathbf{x}$ of $\mathbb{R}^N$ such that

$$\mathbf{x} = \sum_{i=0}^k t_i \mathbf{x}_i, \quad \text{where } \sum_{i=0}^k t_i = 1.$$

It is simple algebra to check that P can also be expressed as the set of all points $\mathbf{x}$ such that

$$(*) \quad \mathbf{x} = \mathbf{x}_0 + \sum_{i=1}^k a_i (\mathbf{x}_i - \mathbf{x}_0)$$

for some scalars $a_1, \dots, a_k$. Thus P can be described not only as “the plane determined by the points $\mathbf{x}_0, \dots, \mathbf{x}_k$,” but also as “the plane passing through the point $\mathbf{x}_0$ parallel to the vectors $\mathbf{x}_1 - \mathbf{x}_0, \dots, \mathbf{x}_k - \mathbf{x}_0$.”

Consider now the homeomorphism $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ defined by the equation $T(\mathbf{x}) = \mathbf{x} - \mathbf{x}_0$. It is called a *translation* of $\mathbb{R}^N$. Expression $(*)$ shows that this map carries the plane P onto the vector subspace V^k of $\mathbb{R}^N$ having as basis the vectors $\mathbf{x}_1 - \mathbf{x}_0, \dots, \mathbf{x}_k - \mathbf{x}_0$. For this reason, we often call P a *k -plane* in $\mathbb{R}^N$.

Two facts follow at once: First, if $k < N$, the k -plane P necessarily has empty interior in $\mathbb{R}^N$ (because V^k does). And second, if $\mathbf{y}$ is any point of $\mathbb{R}^N$ not lying in P , then the set

$$\{\mathbf{x}_0, \dots, \mathbf{x}_k, \mathbf{y}\}$$

is geometrically independent. For if $\mathbf{y} \notin P$, then $T(\mathbf{y}) = \mathbf{y} - \mathbf{x}_0$ is not in V^k . By a standard theorem of linear algebra, the vectors $\{\mathbf{x}_1 - \mathbf{x}_0, \dots, \mathbf{x}_k - \mathbf{x}_0, \mathbf{y} - \mathbf{x}_0\}$ are linearly independent, from which our result follows.

Definition. A set A of points of $\mathbb{R}^N$ is said to be in *general position in $\mathbb{R}^N$* if every subset of A containing $N + 1$ or fewer points is geometrically independent.

In the case of $\mathbb{R}^3$, this is the same as the definition given earlier, as you can check.

Lemma 50.4. *Given a finite set $\{x_1, \dots, x_n\}$ of points of $\mathbb{R}^N$ and given $\delta > 0$, there exists a set $\{y_1, \dots, y_n\}$ of points of $\mathbb{R}^N$ in general position in $\mathbb{R}^N$, such that $|x_i - y_i| < \delta$ for all i .*

Proof. We proceed by induction. Set $y_1 = x_1$. Suppose that we are given $y_1, \dots, y_p$ in general position in $\mathbb{R}^N$. Consider the set of all planes in $\mathbb{R}^N$ determined by subsets of $\{y_1, \dots, y_p\}$ that contain N or fewer elements. Every such subset is geometrically independent and determines a k -plane of $\mathbb{R}^N$ for some $k \leq N-1$. Each of these planes has empty interior in $\mathbb{R}^N$. Because there are only finitely many of them, their union also has empty interior in $\mathbb{R}^N$. (Recall that $\mathbb{R}^N$ is a Baire space.) Choose y_{p+1} to be a point of $\mathbb{R}^N$ within δ of x_{p+1} that does not lie in any of these planes. It follows at once that the set

$$C = \{y_1, \dots, y_p, y_{p+1}\}$$

is in general position in $\mathbb{R}^N$. For let D be any subset of C containing $N+1$ or fewer elements. If D does not contain y_{p+1} , then D is geometrically independent by the induction hypothesis. If D does contain y_{p+1} , then $D - \{y_{p+1}\}$ contains N or fewer points and y_{p+1} is not in the plane determined by these points, by construction. Then as noted above, D is geometrically independent. ■

Theorem 50.5 (The imbedding theorem). *Every compact metrizable space X of topological dimension m can be imbedded in $\mathbb{R}^{2m+1}$.*

Proof. Let $N = 2m + 1$. Let us denote the square metric for $\mathbb{R}^N$ by

$$|x - y| = \max\{|x_i - y_i|; i = 1, \dots, N\}.$$

Then we can use ρ to denote the corresponding sup metric on the space $\mathcal{C}(X, \mathbb{R}^N)$;

$$\rho(f, g) = \sup\{|f(x) - g(x)|, x \in X\}.$$

The space $\mathcal{C}(X, \mathbb{R}^N)$ is complete in the metric ρ , since $\mathbb{R}^N$ is complete in the square metric.

Choose a metric d for the space X ; because X is compact, d is bounded. Given a continuous map $f: X \rightarrow \mathbb{R}^N$, let us define

$$\Delta(f) = \sup\{\text{diam } f^{-1}(\{z\}) \mid z \in f(X)\}.$$

The number $\Delta(f)$ measures how far f “deviates” from being injective; if $\Delta(f) = 0$, each set $f^{-1}(\{z\})$ consists of exactly one point, so f is injective.

Now, given $\epsilon > 0$, define U_ϵ to be the set of all those continuous maps $f: X \rightarrow \mathbb{R}^N$ for which $\Delta(f) < \epsilon$; it consists of all those maps that “deviate” from being injective by less than ϵ . We shall show that U_ϵ is both open and dense in $\mathcal{C}(X, \mathbb{R}^N)$. It follows that the intersection

$$\bigcap_{n \in \mathbb{Z}_+} U_{1/n}$$

is dense in $\mathcal{C}(X, \mathbb{R}^N)$ and is in particular nonempty.

If f is an element of this intersection, then $\Delta(f) < 1/n$ for every n . Therefore, $\Delta(f) = 0$ and f is injective. Because X is compact, f is an imbedding. Thus, the imbedding theorem is proved.

(1) U_ϵ is open in $\mathcal{C}(X, \mathbb{R}^N)$. Given an element f of U_ϵ , we wish to find some ball $B_\rho(f, \delta)$ about f that is contained in U_ϵ . First choose a number b such that $\Delta(f) < b < \epsilon$. Note that if $f(x) = f(y) = z$, then x and y belong to the set $f^{-1}(\{z\})$, so that $d(x, y)$ must be less than b . It follows that if we let A be the following subset of $X \times X$,

$$A = \{x \times y \mid d(x, y) \geq b\},$$

then the function $|f(x) - f(y)|$ is positive on A . Now A is closed in $X \times X$ and therefore compact; hence the function $|f(x) - f(y)|$ has a positive minimum on A . Let

$$\delta = \frac{1}{2} \min\{|f(x) - f(y)|; x \times y \in A\}.$$

We assert that this value of δ will suffice.

Suppose that g is a map such that $\rho(f, g) < \delta$. If $x \times y \in A$, then $|f(x) - f(y)| \geq 2\delta$ by definition; since $g(x)$ and $g(y)$ are within δ of $f(x)$ and $f(y)$, respectively, we must have $|g(x) - g(y)| > 0$. Hence the function $|g(x) - g(y)|$ is positive on A . As a result, if x and y are two points such that $g(x) = g(y)$, then necessarily $d(x, y) < b$. We conclude that $\Delta g \leq b < \epsilon$, as desired.

(2) U_ϵ is dense in $\mathcal{C}(X, \mathbb{R}^N)$. This is the difficult part of the proof. We need to use the analytic geometry of $\mathbb{R}^N$ discussed earlier. Let $f \in \mathcal{C}(X, \mathbb{R}^N)$. Given $\epsilon > 0$ and given $\delta > 0$, we wish to find a function $g \in \mathcal{C}(X, \mathbb{R}^N)$ such that $g \in U_\epsilon$ and $\rho(f, g) < \delta$.

Let us cover X by finitely many open sets $\{U_1, \dots, U_n\}$ such that

- (1) $\text{diam } U_i < \epsilon/2$ in X ,
- (2) $\text{diam } f(U_i) < \delta/2$ in $\mathbb{R}^N$,
- (3) $\{U_1, \dots, U_n\}$ has order $\leq m + 1$.

Let $\{\phi_i\}$ be a partition of unity dominated by $\{U_i\}$ (see §36). For each i , choose a point $x_i \in U_i$. Then choose, for each i , a point $z_i \in \mathbb{R}^N$ such that z_i is within $\delta/2$ of the point $f(x_i)$, and such that the set $\{z_1, \dots, z_n\}$ is in general position in $\mathbb{R}^N$. Finally, define $g : X \rightarrow \mathbb{R}^N$ by the equation

$$g(x) = \sum_{i=1}^n \phi_i(x) z_i.$$

We assert that g is the desired function.

First, we show that $\rho(f, g) < \delta$. Note that

$$g(x) - f(x) = \sum_{i=1}^n \phi_i(x) z_i - \sum_{i=1}^n \phi_i(x) f(x);$$

here we use the fact that $\sum \phi_i(x) = 1$. Then

$$g(x) - f(x) = \sum \phi_i(x)(z_i - f(x_i)) + \sum \phi_i(x)(f(x_i) - f(x)).$$

Now $|z_i - f(x_i)| < \delta/2$ for each i , by choice of the points z_i . And if i is an index such that $\phi_i(x) \neq 0$, then $x \in U_i$; because we have $\text{diam } f(U_i) < \delta/2$, it follows that $|f(x_i) - f(x)| < \delta/2$. Since $\sum \phi_i(x) = 1$, we conclude that $|g(x) - f(x)| < \delta$. Therefore, $\rho(g, f) < \delta$, as desired.

Second, we show that $g \in U_\epsilon$. We shall prove that if $x, y \in X$ and $g(x) = g(y)$, then x and y belong to one of the open sets U_i , so that necessarily $d(x, y) < \epsilon/2$ (since $\text{diam } U_i < \epsilon/2$). As a result, $\Delta(g) \leq \epsilon/2 < \epsilon$, as desired.

So suppose $g(x) = g(y)$. Then

$$\sum_{i=1}^n [\phi_i(x) - \phi_i(y)]z_i = 0.$$

Because the covering $\{U_i\}$ has order at most $m+1$, at most $m+1$ of the numbers $\phi_i(x)$ are nonzero, and at most $m+1$ of the numbers $\phi_i(y)$ are nonzero. Thus, the sum $\sum [\phi_i(x) - \phi_i(y)]z_i$ has at most $2m+2$ nonzero terms. Note that the sum of the coefficients vanishes because

$$\sum [\phi_i(x) - \phi_i(y)] = 1 - 1 = 0.$$

The points z_i are in general position in $\mathbb{R}^N$, so that any subset of them having $N+1$ or fewer elements is geometrically independent. And by hypothesis $N+1 = 2m+2$. (Aha!) Therefore, we conclude that

$$\phi_i(x) - \phi_i(y) = 0$$

for all i .

Now $\phi_i(x) > 0$ for some i , so that $x \in U_i$. Since $\phi_i(y) = \phi_i(x)$, we have $y \in U_i$ also, as asserted. ■

To give some content to the imbedding theorem, we need some more examples of spaces that are finite dimensional. We prove the following theorem.

Theorem 50.6. Every compact subspace of $\mathbb{R}^N$ has topological dimension at most N .

Proof The proof is a generalization of the proof given in Example 3 for $\mathbb{R}^2$. Let ρ be the square metric on $\mathbb{R}^N$.

Step 1. We begin by breaking $\mathbb{R}^N$ up into "unit cubes." Define $\mathcal{J}$ to be the following collection of open intervals in $\mathbb{R}$:

$$\mathcal{J} = \{(n, n+1) \mid n \in \mathbb{Z}\},$$

and define $\mathcal{K}$ to be the following collection of one-point sets in $\mathbb{R}$:

$$\mathcal{K} = \{\{n\} \mid n \in \mathbb{Z}\}$$

If M is an integer such that $0 \leq M \leq N$, let $\mathcal{C}_M$ denote the set of all products

$$C = A_1 \times A_2 \times \cdots \times A_N,$$

where exactly M of the sets A_i belong to $\mathcal{J}$, and the remainder belong to $\mathcal{K}$. If $M > 0$, then C is homeomorphic to the product $(0, 1)^M$ and will be called an **M -cube**. If $M = 0$, then C consists of a single point and will be called a **0-cube**.

Let $\mathcal{C} = \mathcal{C}_0 \cup \mathcal{C}_1 \cup \cdots \cup \mathcal{C}_N$. Note that each point $\mathbf{x}$ of $\mathbb{R}^N$ lies in precisely one element of $\mathcal{C}$ because each real number x_i lies in precisely one element of $\mathcal{J} \cup \mathcal{K}$. We shall expand each element C of $\mathcal{C}$ slightly to an open set $U(C)$ of $\mathbb{R}^N$ of diameter at most $3/2$, in such a way that if C and D are two different M -cubes, then $U(C)$ and $U(D)$ are disjoint.

Let $\mathbf{x} = (x_1, \dots, x_N)$ be a point of the M -cube C . We show that there is a number $\epsilon(\mathbf{x}) > 0$ such that the $\epsilon(\mathbf{x})$ -neighborhood of $\mathbf{x}$ intersects no M -cube other than C . If C is a 0-cube, we set $\epsilon(\mathbf{x}) = 1/2$ and we are finished. Otherwise, $M > 0$, and exactly M of the numbers x_i are not integers. Choose $\epsilon \leq 1/2$ so that for each x_i that is not an integer, the interval $(x_i - \epsilon, x_i + \epsilon)$ contains no integer. If $\mathbf{y} = (y_1, \dots, y_N)$ is a point lying in the ϵ -neighborhood of $\mathbf{x}$, then y_i is nonintegral whenever x_i is nonintegral. This means that $\mathbf{y}$ either belongs to the same M -cube as $\mathbf{x}$ does, or $\mathbf{y}$ belongs to some L -cube for $L > M$. In either case, the ϵ -neighborhood of $\mathbf{x}$ intersects no M -cube other than C .

Given an M -cube C , we define the neighborhood $U(C)$ of C to be the union of the $\epsilon(\mathbf{x})/2$ -neighborhoods of $\mathbf{x}$ for all $\mathbf{x} \in C$. It is then immediate that if C and D are different M -cubes, $U(C)$ and $U(D)$ are disjoint. Furthermore, if $\mathbf{z}$ is a point of $U(C)$, then $d(\mathbf{z}, \mathbf{x}) < \epsilon(\mathbf{x})/2 < 1/4$ for some point $\mathbf{x}$ of C . Since C has diameter 1, the set $U(C)$ has diameter at most $3/2$.

Step 2. Given M with $0 \leq M \leq N$, define $\mathcal{A}_M$ to be the collection of all sets $U(C)$, where $C \in \mathcal{C}_M$. The elements of $\mathcal{A}_M$ are disjoint, and each has diameter at most $3/2$. The remainder of the proof is a copy of the proof given in Example 3 for $\mathbb{R}^2$. ■

Corollary 50.7. Every compact m -manifold has topological dimension at most m .

Corollary 50.8. Every compact m -manifold can be imbedded in $\mathbb{R}^{2m+1}$.

Corollary 50.9. Let X be a compact metrizable space. Then X can be imbedded in some euclidean space $\mathbb{R}^N$ if and only if X has finite topological dimension.

As mentioned earlier, much of what we have proved holds without assumption of compactness. We ask you to prove the appropriate generalizations in the exercises that follow.

One thing we do *not* ask you to prove is the fact that the topological dimension of an m -manifold is precisely m . And for good reason; the proof requires the tools of algebraic topology.

Nor do we ask you to prove that $N = 2m + 1$ is the smallest value of N such that every compact metrizable space of topological dimension m can be imbedded in $\mathbb{R}^N$. The reason is the same. Even in the case of a linear graph, where $m = 1$, the proof is nontrivial, as we remarked earlier.

For further results in dimension theory, the reader is referred to the classical book of Hurewicz and Wallman [H-W]. In particular, this book discusses another, entirely different, definition of topological dimension, due to Menger and Urysohn. It is an inductive definition. The empty set has dimension -1 . And a space has dimension at most n if there is a basis for its topology such that for each basis element B , the boundary of B has dimension at most $n - 1$. The dimension of a space is the smallest value of n for which this condition holds. This notion of dimension agrees with ours for compact metrizable spaces.

Exercises

1. Show that any discrete space has dimension 0.
2. Show that any connected T_1 space having more than one point has dimension at least 1.
3. Show that the topologist's sine curve has dimension 1.
4. Show that the points $0, \epsilon_1, \epsilon_2, \epsilon_3$, and $(1, 1, 1)$ are in general position in $\mathbb{R}^3$. Sketch the corresponding imbedding into $\mathbb{R}^3$ of the complete graph on five vertices.
5. Examine the proof of the imbedding theorem in the case $m = 1$ and show that the map g of part (2) actually maps X onto a linear graph in $\mathbb{R}^3$.
6. Prove the following:

Theorem. Let X be a locally compact Hausdorff space with a countable basis, such that every compact subspace of X has topological dimension at most m . Then X is homeomorphic to a closed subspace of $\mathbb{R}^{2m+1}$.

Proof. If $f : X \rightarrow \mathbb{R}^N$ is a continuous map, we say $f(x) \rightarrow \infty$ as $x \rightarrow \infty$ if given n , there is a compact subspace C of X such that $f(x) > n$ for $x \in X - C$.

- (a) Let $\bar{\rho}$ be the uniform metric on $\mathcal{C}(X, \mathbb{R}^N)$. Show that if $f(x) \rightarrow \infty$ as $x \rightarrow \infty$ and $\bar{\rho}(f, g) < 1$, then $g(x) \rightarrow \infty$ as $x \rightarrow \infty$.
- (b) Show that if $f(x) \rightarrow \infty$ as $x \rightarrow \infty$, then f extends to a continuous map of one-point compactifications. Conclude that if f is injective as well, then f is a homeomorphism of X with a closed subspace of $\mathbb{R}^N$.
- (c) Given $f : X \rightarrow \mathbb{R}^N$ and given a compact subspace C of X , let

$$U_\epsilon(C) = \{f \mid \Delta(f|C) < \epsilon\}.$$

Show that $U_\epsilon(C)$ is open in $\mathcal{C}(X, \mathbb{R}^N)$.

- (d) Show that if $N = 2m + 1$, then $U_\epsilon(C)$ is dense in $\mathcal{C}(X, \mathbb{R}^N)$. [Hint: Given f and given $\epsilon, \delta > 0$, choose $g : C \rightarrow \mathbb{R}^N$ so that $d(f(x), g(x)) < \delta$ for

$x \in C$, and $\Delta(g) < \epsilon$. Extend $f - g$ to $h : X \rightarrow [-\delta, \delta]^N$ using the Tietze theorem.]

- (e) Show there exists a map $f : X \rightarrow \mathbb{R}$ such that $f(x) \rightarrow \infty$ as $x \rightarrow \infty$. [Hint: Write X as the union of compact subspaces C_n such that $C_n \subset \text{Int } C_{n+1}$ for each n .]
 (f) Let C_n be as in (e). Use the fact that $\bigcap U_{1/n}(C_n)$ is dense in $\mathcal{C}(X, \mathbb{R}^N)$ to complete the proof.

7. Corollary. Every m -manifold can be imbedded in $\mathbb{R}^{2m+1}$ as a closed subspace.

8. Recall that X is said to be σ -compact if there is a countable collection of compact subspaces of X whose interiors cover X .

Theorem. Let X be a σ -compact Hausdorff space. If every compact subspace of X has topological dimension at most m , then so does X .

Proof. Let $\mathcal{A}$ be an open cover of X . Find an open cover $\mathcal{B}$ of X refining $\mathcal{A}$ that has order at most $m + 1$, as follows:

- (a) Show that $X = \bigcup X_n$, where X_n is compact and $X_n \subset \text{Int } X_{n+1}$ for each n . Let $X_0 = \emptyset$.
 (b) Find an open covering $\mathcal{B}_0$ of X refining $\mathcal{A}$ such that for each n , each element of $\mathcal{B}_0$ that intersects X_n lies in X_{n+1} .
 (c) Suppose $n \geq 0$ and $\mathcal{B}_n$ is an open covering of X refining $\mathcal{B}_0$ such that $\mathcal{B}_n$ has order at most $m + 1$ at points of X_n . Choose an open covering $\mathcal{C}$ of X refining $\mathcal{B}_n$ that has order at most $m + 1$ at points of X_{n+1} . Choose $f : \mathcal{C} \rightarrow \mathcal{B}_n$ so that $C \subset f(C)$. For $B \in \mathcal{B}_n$, let $D(B)$ be the union of those C for which $f(C) = B$. Let $\mathcal{B}_{n+1}$ consist of all sets $B \in \mathcal{B}_n$ for which $B \cap X_{n-1} \neq \emptyset$, along with all sets $D(B)$ for which $B \in \mathcal{B}_n$ and $B \cap X_{n-1} = \emptyset$. Show that $\mathcal{B}_{n+1}$ is an open covering of X that refines $\mathcal{B}_n$ and has order at most $m + 1$ at points of X_{n+1} .
 (d) Define $\mathcal{B}$ as follows: Given a set B , it belongs to $\mathcal{B}$ if there is an N such that $B \in \mathcal{B}_n$ for all $n \geq N$.

9. Corollary. Every m -manifold has topological dimension at most m .

10. Corollary. Every closed subspace of $\mathbb{R}^N$ has topological dimension at most N .

11. Corollary. A space X can be imbedded as a closed subspace of $\mathbb{R}^N$ for some N if and only if X is locally compact and Hausdorff with a countable basis, and has finite topological dimension.

*Supplementary Exercises: Locally Euclidean Spaces

A space X is said to be **locally m -euclidean** if for each $x \in X$, there is a neighborhood of x that is homeomorphic to an open set of $\mathbb{R}^m$. Such a space X automatically satisfies the T_1 axiom, but it need not be Hausdorff. (See the exercises of §36.) However, if X is Hausdorff and has a countable basis, then X is called an **m -manifold**.

Throughout these exercises, let X be a space that is locally m -euclidean.

1. Show that X is locally compact and locally metrizable.
2. Consider the following conditions on X :
 - (i) X is compact Hausdorff.
 - (ii) X is an m -manifold.
 - (iii) X is metrizable.
 - (iv) X is normal.
 - (v) X is Hausdorff.
 Show that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (v).
3. Show that $\mathbb{R}$ is locally 1-euclidean and satisfies (ii) but not (i).
4. Show that $\mathbb{R} \times \mathbb{R}$ in the dictionary order topology is locally 1-euclidean and satisfies (iii) but not (ii).
5. Show that the long line is locally 1-euclidean and satisfies (iv) but not (iii). (See the exercises of §24.)
- *6. There is a space that is locally 2-euclidean and satisfies (v) but not (iv). It is constructed as follows. Let A be the following subspace of $\mathbb{R}^3$:

$$A = \{(x, y, 0) \mid x > 0\}.$$

Given c real, let B_c be the following subspace of $\mathbb{R}^3$:

$$B_c = \{(x, y, c) \mid x \leq 0\}.$$

Let X be the set that is the union of A and all the spaces B_c , for c real. Topologize X by taking as a basis all sets of the following three types:

- (i) U , where U is open in A .
- (ii) V , where V is open in the subspace of B_c consisting of points with $x < 0$.
- (iii) For each open interval $I = (a, b)$ of $\mathbb{R}$, each real number c , and each $\epsilon > 0$, the set $A_c(I, \epsilon) \cup B_c(I, \epsilon)$, where

$$A_c(I, \epsilon) = \{(x, y, 0) \mid 0 < x < \epsilon \text{ and } c + ax < y < c + bx\},$$

$$B_c(I, \epsilon) = \{(x, y, c) \mid -\epsilon < x \leq 0 \text{ and } a < y < b\}.$$

The space X is called the "Prüfer manifold."

- (a) Sketch the sets $A_c(I, \epsilon)$ and $B_c(I, \epsilon)$.
- (b) Show the sets of types (i)–(iii) form a basis for a topology on X .
- (c) Show the map $f_c : \mathbb{R}^2 \rightarrow X$ given by

$$f_c(x, y) = \begin{cases} (x, c + xy, 0) & \text{for } x > 0, \\ (x, y, c) & \text{for } x \leq 0 \end{cases}$$

defines a homeomorphism of $\mathbb{R}^2$ with the subspace $A \cup B_c$ of X .

- (d) Show that $A \cup B_c$ is open in X ; conclude that X is 2-euclidean.
- (e) Show that X is Hausdorff.
- (f) Show that X is not normal. [*Hint*: The subspace

$$L = \{(0, 0, c) \mid c \in \mathbb{R}\}$$

of X is closed and discrete. Compare Example 3 of §31.]

- 7. Show that X is Hausdorff if and only if X is completely regular.
- 8. Show that X is metrizable if and only if X is paracompact Hausdorff.
- 9. Show that if X is metrizable, then each component of X is an m -manifold.